

Algorithms for Lattice Fermions User Workshop 2020

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Pre-conference event: Installation
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ALF

Installation notes

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Table of contents

1. Introduction

- The ALF project

- Prerequisites

2. ALF Installation

- Installing ALF prerequisites

- Installing ALF

- Testing ALF

3. pyALF installation

- Installing pyALF prerequisites

- Downloading and testing pyALF

Introduction

The ALF project contains the following public parts:

- **ALF**: our main package, responsible for performing auxiliary-field quantum Monte Carlo simulations; also its **documentation** can be found here
- **pyALF**: provides a Python interface and scripts, as well as Jupyter notebooks
- **ALF_Tutorial**: stores ALF's tutorial, including solutions to the proposed exercises
- **ALF_Presentations**: where, for instance, these notes are made available
- **ALF_code**: the main package's legacy version, ALF 1.0

Version-control is done using Git and they are available in the GitLab repository:

<https://git.physik.uni-wuerzburg.de/ALF>.

These notes cover the installation for ALF and pyALF.

To run ALF and pyALF you need the following installed in your machine:

- For ALF:
 - a Fortran compiler, such as `gfortran` or `ifort`
 - Blas library
 - Lapack library
- For pyALF
 - Python3 with
 - SciPy
 - NumPy
 - matplotlib
 - Jupyter

ALF Installation

Linux

To install the required libraries and compiler in one of the following Linux distributions, enter in your terminal:

- Debian/Ubuntu/Linux Mint:
`sudo apt-get install gfortran liblapack-dev make`
- Red Hat/Fedora/CentOS:
`sudo yum install gcc-gfortran make liblapack-devel`
- OpenSuSE/SLES:
`sudo zypper install gcc-gfortran make lapack-devel`
- Arch Linux:
`pacman -S make gcc-fortran lapack`

MacOS

For MacOS, Xcode and the Apple developer tools should be installed.

gfortran for MacOS can be found at:

<https://gcc.gnu.org/wiki/GFortranBinaries#MacOS>,

where detailed information on how to install the package can be found.

Windows 10

For Windows 10, we suggest using the Windows subsystem for Linux to install Ubuntu and then follow the instructions for Ubuntu. For installation instructions see:

<https://docs.microsoft.com/en-us/windows/wsl/install-win10>

Older Windows versions

For older Windows versions, you can use Cygwin, found at <https://www.cygwin.com/>, which provides a Unix-like environment for Windows.

Installing ALF

With its dependencies in place, we can proceed to the proper ALF installation.

ALF's source code can be downloaded by entering the following in a shell:

```
git clone https://git.physik.uni-wuerzburg.de/ALF/ALF.git
```

To set the necessary environment variables and compiler directives run the script `configure.sh`:

```
source configure.sh [MACHINE] [MODE] [STAB]
```

where `MACHINE` can be, e.g., `GNU`, for choosing `gfortran` (see ALF's documentation).

In order to compile the libraries, the analysis routines and the QMC program at once, just execute the single command:

```
make
```

When the compilation is over you are in position to run simulations with ALF.

Now let us use ALF:

1. Create a data directory with the content of `Start` found in the folder `Scripts_and_Parameters_files`:

```
cp -r $ALF_DIR/Scripts_and_Parameters_files/Start ./Run && cd ./Run/
```

2. Run an executable Hamiltonian, e.g.:

```
$ALF_DIR/Prog/Hubbard_Plain_Vanilla.out
```

3. Perform default analysis¹:

```
$ALF_DIR/Analysis/ana.out *
```

4. Check results. For instance, `Ener_scalJ` should be around -30 for default parameters.

¹The `analysis.sh` bash script from earlier versions of ALF, run without arguments, is still available in the `Start` directory.

pyALF installation

Installing pyALF prerequisites

You can either manually install the python packages through the package manager of your operation system, or having all in one using Anaconda. For Windows you should use the former through the Linux subsystem.

Direct installation For Windows (Ubuntu/Debian on Linux subsystem), or if you don't want to install Anaconda, you can install the needed python packages with:

```
sudo apt-get install jupyter-notebook python3 python3-numpy python3-scipy python3-matplotlib
```

For other distributions, similar packages are available.

Anaconda

All pyALF's key dependencies – Python, Jupyter and their packages – can be easily installed on a variety of platforms by using Anaconda.

Detailed installation instructions and the necessary files can be found at <https://docs.anaconda.com/anaconda/install/>.

Downloading and testing pyALF

With pyALF's dependencies taken care of, now only the package itself is missing.

The package can be downloaded by entering the following in a shell:

```
git clone https://git.physik.uni-wuerzburg.de/ALF/pyALF.git
```

Finally, append pyALF's path to the environment variable PYTHONPATH by, e.g., adding to `.bashrc`:

```
export PYTHONPATH="/local/path/to/pyALF:$PYTHONPATH"
export ALF_DIR="/local/path/to/ALF"
```

So that python can find the `py_alf` module.

pyALF contains, besides the main script `py_alf.py` which provides ALF's Python interface, scripts and Jupyter notebooks we can use for testing. For instance:

```
python3 Hubbard_Plain_Vanilla.py
```

should output a file named `Ener_Plain_Vanilla.dat`, containing the measured energy of the system, which should be approximately -2.10 for the default parameters.